THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM ONE
031 PHYSICS
MARKING SCHEME AUGUST 2026

SECTION A (15 MARKS)

1. **@01 Mark = 10 marks**

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	D	B	A	C	A	B	B	A	C

2. **@01 Mark = 05 marks**

i	ii	Iii	iv	v
C	A	D	E	B

SECTION B (MARKS 70)

3. (a) Problem identification: Is a first step in scientific investigation, or Problem identification is the process of recognizing and clearly defining a specific challenge or question that needs to be addressed through scientific investigation or experimentation 04 marks

(b)

Observe patterns in the ball's behavior, such as where it consistently stops. Consider variables like ramp angle, friction, or ball surface. The problem might be due to surface resistance or lack of sufficient force to keep the ball moving. 06marks

4. (a) A hypothesis is a testable prediction or educated guess about the relationship between two or more variables, based on observations and prior knowledge Or is an intelligent guess 04 marks

(b) If a balloon is rubbed against hair, then it will become negatively charged and attract small, neutral objects like paper due to electrostatic force. 06marks

5. (a) Any three:

(i) Changes the state of motion of an object.
(ii) Changes the direction of motion.
(iii) Changes the speed/velocity of an object.
(iv) Changes the shape or size of an object.
(v) Causes an object to rotate/turn.
(3 marks: 1 mark each)
(b)
(i)
Extension = Final length − Original length
= 26 cm − 20 cm
= **6 cm. (2 marks)**
(ii) **Elasticity. (1 mark)**
(iii) Any two:
(i) Magnitude of the applied force.

(ii) Nature/type of material.
(iii) Original length/dimensions of the material.
(2 marks: 1 mark each)
(c) The material develops an **elastic restoring force** that tends to return it to its original shape and size when the applied force is removed, as long as its elastic limit is not exceeded. **(2 marks)**

6. (a) Factors Affecting Solid Pressure (Total: 04 Marks)

Factor 1: The acting Force / Weight (F): Pressure is directly proportional to the perpendicular force applied. A larger force increases pressure. (02 Marks)

Factor 2: The Surface Area in contact (A): Pressure is inversely proportional to the area. A smaller contact area increases pressure. (02 Marks)

(b) Linear Motion Calculations (Total: 06 Marks)

Given Parameters:Initial velocity (u) = 0 m/s (since it starts from rest)

Final velocity (v) = 20 m/sTime (t) = 5 s

(i)Calculate the Acceleration (a) (03 Marks)

$$\text{Acceleration\ } (a)=\frac{v-u}{t} \text{ [Formula: 01 Mark]}$$

$$a=\frac{20\text{\ m/s}-0\text{\ m/s}}{5\text{\ s}} \text{ [Substitution:\ 01\ Mark]}$$

$$a=\mathbf{4}\text{\ m/s}^2 \text{ [Answer\ with\ correct\ unit:\ 01\ Mark]}$$

(ii) Calculate the Total Distance Covered (s) (03 Marks)

Using the second equation of motion:

$$s=ut+\frac{1}{2}at^2 \text{ [Formula:\ 01\ Mark]}$$

$$s=(0\times5)+\frac{1}{2}(4)(5)^2 \text{ [Substitution: 01 Mark]}$$

$$s=0+2(25)=50 \textit{ meters } (m)[\textit{Answer wit}\text{h } \textit{correct unit}:01\textit{Mark}]$$

(Alternative Method using

$$s=\frac{(u+v)}{2}\times t \Longrightarrow \frac{(0+20)}{2}\times5=10\times5$$

=50 m is awarded full marks).

7.(a) To calculate the accurate diameter of the rod using the screw gauge, we need to consider the contribution of both the sleeve and the thimble, taking into the account the least count of the instrument 04 marks
Sleeve shows the main scale reading in mm.

Thimble shows finer readings in hundredths of a mm.

Least count (0.01 mm) ensures high precision.

To ensure accuracy, avoid zero error, calibrate properly, and take multiple readings.

(b)

Sleeve reading = 1.6 cm = 16.00 mm 01marks

Thimble reading = 48 × 0.01 mm = 0.48 mm 02marks

Total diameter = 16.00 + 0.48 = 16.48 mm 02 marks

So the correct reading is 16.48mm 01marks

8.(a) Force is a physical quantity that causes an object to move, stop, or change direction; it is a push or pull on an object. 02marks

(b) List four naturally occurring forces

i. Gravitational force ……02marks

ii. Electrostatic force ……02marks

iii. strong force ……02marks

iv. weak force ……02marks

9.Four differences between mass and weight

Mass	Weight
i. Measure of matter	i. Measure of gravitational pull
ii. Constant everywhere	ii. Changes with location (gravity)
iii. Measured in kilograms (kg)	iii. Measured in newton's (N)
iv. Scalar quantity	iv. Vector quantity
v. Measured by balance	v. Measured by spring balance

2marks @

SECTION C (15 MARKS)

10.

(a) Identification of Forces (Total: 03 Marks)i. Two bar magnets push each other away.

(i) Magnetic Force / Repulsive Magnetic ForceAward: 01 Markii. A book sliding across a table slowly comes to a stop.

(ii) Frictional Force / FrictionAward: 01 Markiii. A dropped ball falls towards the ground.

(iii) Gravitational Force / Gravity / WeightAward: 01 Mark10.

(b) Branches of Physics Explained (Total: 04 Marks)Any four correct responses among the following standard branches

(i) Mechanics: The branch of physics dealing with the motion of objects under the influence of forces.

(ii) Thermodynamics: The branch of physics that studies heat, temperature, and their relation to work and energy.

(iii) Optics: The study of the behavior and properties of light, including its interactions with matter.

(iv) Electromagnetism: The branch that investigates electrical charges, magnetic fields, and electromagnetic waves.

(v) Astrophysics: The study of the physical properties and celestial phenomena of stars, planets, and the universe.

(vi) Nuclear Physics: The branch concerned with the structure, behavior, and components of atomic nuclei.

(c)

Volume of block = 200 cm^3 = 2×10^{-4} m^3. (1 mark)
Mass of water displaced:
$m=\rho V = 1000 \times 2 \times 10-4 = 0.2kg$
(2 marks)
Weight of displaced water:
$W=mg=0.2 \times 10=2N$
Thus, the upthrust = 2 N. (1 mark)
Since the upthrust is equal to the weight of the displaced water, the block floats/neither continues sinking nor rises when in equilibrium. (1 mark)